Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: *MASSIVE* | ***Context:*** *Ethiopian Intercity Travelers — Amharic-Language Booking Bot*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output label space (60 intents and 55 slots [Q1]) is the same structure used to generate the input ontology, so the same gaps apply on the output side. Operator-specific slot types required by the deployment — operator_name, seat_class, route_closure_reason, surcharge_event, trip_change_type, departure_terminal — are absent from the 55-slot taxonomy (regional context, operator_specific_slot_types). The paper itself describes the schema as 'relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options' [Q108]. Output decision rules (label/F1/exact match) are language-agnostic, so cultural decision-rule misalignment per se is limited — the violation is primarily content-validity (missing categories) rather than structural-validity for the labels that do exist. The text-to-text decoder format ('sequence of labels ... followed by the intent' [Q73]) is compatible with the deployment.

ID	Evidence
Q1	“MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.” (p.1)
Q73	“The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.” (p.7)
Q108	“Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options.” (p.9)

Strengths

- 55-slot taxonomy includes generic temporal, location, and entity slots [Q1] that partially cover booking attributes
- Slot localization metadata (translate/localize/unchanged) is preserved [Q58], allowing post-hoc filtering
- Judgment metadata for intent/slot semantic match is included [Q63, Q64], supporting label-quality auditing

ID	Evidence
Q1	“MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.” (p.1)
Q58	“The metadata of the released dataset includes whether the worker elected to "localize," "translate," or keep the slot "unchanged," primarily for the purposes of researchers evaluating machine translation systems, where it would be unreasonable to expect the system to "localize" to a specific song name the worker selected.” (p.6)
Q63	“The output of the second workflow (the fully localized utterance) is judged by three workers for (1) whether the utterance matches the intent semantically, (2) whether the slots match their labels semantically, (3) grammaticality and naturalness, (4) spelling, and (5) language identification—English or mixed utterances are acceptable if that is natural for the language, but localizations without any tokens in the target language were not accepted.” (p.6)
Q64	“These judgments are also included in the metadata of the dataset.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question

OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Generic booking-relevant slot types (date, time, place_name, person, etc.) implied by 18 domains [Q1] partially apply. Ethiopian-specific slots (operator_name, seat_class, departure_terminal) are absent — confirmed by gap_id 11 (operator-specific seat-class taxonomy not publicly available).

OO-2: Missing categories: operator_name, seat_class, route_closure_reason, surcharge_event (holiday-driven), trip_change_type, departure_terminal — none are in the 55-slot taxonomy [Q1, Q108].

OO-3: The 60 intents reflect global virtual-assistant priorities [Q16] (smart-home, music, weather, etc.) that encode non-Ethiopian commercial assumptions; this is more construct-irrelevant variance than a values-encoding violation.

OO-4: Stakeholder-driven taxonomy redesign is needed for operator-specific slots; gap_id 11 confirms no public operator taxonomy exists, requiring direct elicitation.

OO-5: Structural validity is only mildly affected (label format is compatible); content validity is moderately to strongly affected by missing slot types [Q108].

ID	Evidence
Q1	“MASSIVE contains 1M realistic, parallel, labeled virtual assistant utterances spanning 51 languages, 18 domains, 60 intents, and 55 slots.” (p.1)
Q16	“Second, we determined existing languages available in major virtual assistants, such that the dataset could be used to benchmark today's systems.” (p.3)
Q73	“The decoder output is a sequence of labels (including the Other label) for all of the tokens followed by the intent.” (p.7)
Q108	“Fourth, our labeling schema is relatively simple when compared with hierarchical labeling schemata or flat schemata with more intent and slot options.” (p.9)
WEB-21	No African transport-domain slot taxonomy exists — operator-specific slot types must be developed locally (https://aclanthology.org/2024.lrec-main.561/)

Information Gaps

- Detailed list of MASSIVE's 55 slot types to determine partial overlap with booking-domain needs

Requires Expert Verification

- Operator-specific seat-class enumeration and route-closure reason categories (gap_id 11)

Remediation

Gap 1: Operator-specific slot types (operator_name, seat_class, route_closure_reason, surcharge_event, trip_change_type, departure_terminal) missing from 55-slot taxonomy

Recommendation: Define an extended slot taxonomy with operator-specific types and an enumerated value set per slot (e.g., seat_class ∈ {VIP, economy, sleeper, …} resolved via direct operator elicitation). Annotate the supplementary test set against this extended taxonomy, keeping MASSIVE-compatible slot types for backward comparability.